

DESF5 - Desafio Final

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PASSO 1 — Criar um usuário IAM:

Console AWS → Pesquise por IAM → Users → Criar usuário → Dê nome para o usuário → “Próximo” → Determine as políticas → “Próximo” → Revise as informações → “Criar usuário”.

Nome do usuário

kafka-spark-cesteves

The screenshot shows the 'Specify user details' step in the AWS IAM console. On the left, a progress bar indicates three steps: 'Etapas 1 Especificar detalhes do usuário' (selected), 'Etapas 2 Definir permissões', and 'Etapas 3 Revisar e criar'. The main area is titled 'Especificar detalhes do usuário'. Under 'Detalhes do usuário', the 'Nome do usuário' field contains 'kafka-spark-cesteves'. A note below states: 'O nome de usuário pode ter até 64 caracteres. Caracteres válidos: A-Z, a-z, 0-9, and +, -, @, _ (if not)'. There is an unchecked checkbox for 'Fornecer acesso para os usuários ao Console de Gerenciamento da AWS - opcional!'. A blue box at the bottom contains a tip: 'Se você estiver criando acesso programático por meio de chaves de acesso ou credenciais específicas de serviço para o AWS CodeCommit ou o Amazon Keyspaces, poderá gerá-las depois de criar esse usuário do IAM. Saiba mais'. At the bottom right are 'Cancelar' and 'Próximo' buttons.

Permissões: amazonSESfull

The screenshot shows the 'Define permissions' step in the AWS IAM console. The progress bar on the left shows 'Etapas 1 Especificar detalhes do usuário' and 'Etapas 2 Definir permissões' (selected). The main area is titled 'Definir permissões' with a subtitle: 'Adicione usuário a um grupo existente ou crie um novo. Usar grupos é uma prática recomendada para gerenciar as permissões do usuário por funções de trabalho. Saiba mais'. Under 'Opções de permissões', there are three radio buttons: 'Adicionar usuário ao grupo' (selected), 'Copiar permissões', and 'Anexar políticas diretamente'. The 'Anexar políticas diretamente' option has a blue box with a tip: 'Anexe uma política gerenciada diretamente a um usuário. Como prática recomendada, recomendamos anexar políticas a um grupo. Em seguida, adicione o usuário ao grupo apropriado.' Below this is a section 'Políticas de permissões (1/1443)' with a search bar containing 'amazonSESfull' and a filter dropdown set to 'Todos os tipos'. A table lists the search results: one result for 'AmazonSESFullAccess' (Gerenciadas pela AWS) with 0 associated entities. At the bottom right are 'Cancelar', 'Anterior', and 'Próximo' buttons.

PASSO 2 — Criar as credenciais de acesso:

Acessar o usuário criado → Aba: Credenciais de segurança → “Criar chave de acesso” → Selecione “Aplicação Executada fora da AWS” → “Próximo” → “Criar chave de acesso” → Armazene os dados **Chave de acesso** e **Chave de acesso secreta** → Concluído.

Criar chave de acesso

Chave de acesso: **AKIASL....**

Chave de acesso secreta: **IUnm9...**

Chave salva em: L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD

Etapa 1

Práticas recomendadas e alternativas para chaves de acesso

Etapa 2 - opcional

Definir etiqueta de descrição

Etapa 3

Recuperar chaves de acesso

Recuperar chaves de acesso

Informações

Chave de acesso

Se você perder ou esquecer sua chave de acesso secreta, não poderá recuperá-la. Em vez disso, crie uma nova chave de acesso e torne a chave antiga inativa.

Chave de acesso

Chave de acesso secreta

AKIASLRKSVSPADKUSMFY

IUnm9wvFewanUBgJsdSa2Lpf8TN/ET1PVZVjQX

Ocultar

Acesse as principais práticas recomendadas

- Nunca armazene sua chave de acesso em texto sem formatação, em um repositório de códigos ou em código.
- Desabilite ou exclua a chave de acesso quando não for mais necessária.
- Habilite permissões de menor privilégio.
- Altere regularmente as chaves de acesso.

Para obter mais detalhes sobre o gerenciamento de chaves, consulte as [Práticas recomendadas para o gerenciamento de chaves da AWS](#).

Baixar arquivo .csv

Concluído

Login do console

Link de login do console: <https://162263641246.signin.aws.amazon.com/console>

Identity and Access Management (IAM)

Gerenciamento de acesso

Grupos de usuários

Usuários

Funções

Políticas

Procedimentos de identidade

Configurações da conta

Gerenciamento de acesso MFA

Solicitações de delegação

Integração de SSO

Relatório de acesso

Assess de acesso

Análise de recursos

Acesso não utilizado

Configurações de Auditoria

Relatório de credenciais

Atividade da organização

Políticas de controle de acesso

Políticas de controle de recursos

Central de Identidades da AWS

AWS Organizations

kafka-spark-cesteves

Excluir

Resumo

ARN

am:iam::162263641246:user/kafka-spark-cesteves

Acesso ao console

Desabilitado

Chave de acesso 1

AKIASLRKSVSPADKUSMFY - Active

Nome criado. Crie uma nova.

Chave de acesso 2

Crie uma nova chave de acesso

Permissões

Grupos

Etiquetas

Credenciais de segurança

Último acesso

Login do console

Link de login de console

https://162263641246.signin.aws.amazon.com/console

Senha do console

Não habilitado

Habilitar acesso ao console

Autenticação multifator (MFA) (0)

Use a MFA para aumentar a segurança do seu ambiente da AWS. Para logar com a MFA, você precisa de um dispositivo MFA. Cada usuário pode ter no máximo dois dispositivos MFA vinculados.

Remover

Sincronizar novamente

Atribuir dispositivo com MFA

Tipos

Identificador

Certificações

Criado em

Nenhum dispositivo MFA. Adicione um dispositivo MFA para melhorar a segurança do seu ambiente da AWS.

Atribuir dispositivo com MFA

Chaves de acesso (1)

Use chaves de acesso para fazer chamadas programáticas para a AWS da AWS CLI, do AWS Tools for PowerShell, do SDKs da AWS ou de chamadas de API diretas da AWS. Você pode ter no máximo duas chaves de acesso (ativas ou inativas) por usuário.

Crie uma nova chave de acesso

AKIASLRKSVSPADKUSMFY

Descrição

Projeto Desafio Final Pós-Engenharia e Arquitetura de Dados com IA

Status

Active

Última utilização

hazum

Criado

2 minutos atrás

Última região utilizada

us-east-1

Último serviço utilizado

N/A

Ações

PASSO 3 — Criar o Bucket S3 Bronze (Data Lake):

Console AWS → Pesquise por S3 → “buckets” → “Criar bucket” → Dê nome ao bucket → Criar bucket.

Atenção: Tudo que conversa com o S3 precisa apontar para a MESMA região do bucket, neste caso: América do Sul (São Paulo) sa-east-1.

1. Buckets: desafio-bronze-cesteves

Bucket criado com êxito "desafio-bronze-cesteves"

Para fazer upload de arquivos e pastas ou definir configurações adicionais de bucket, selecione Visualizar detalhes.

Visualizar detalhes

Buckets de uso geral

Todas as regiões da AWS

Buckets de diretórios

Buckets de uso geral (2)

Informações

Copiar ARN

Vazio

Excluir

Criar bucket

Buckets são contêineres para dados armazenados no S3.

Encontrar buckets por nome

< 1 >

Nome

Região da AWS

Data de criação

desafio-bronze-cesteves

América do Sul (São Paulo) sa-east-1

3 Feb 2026 02:45:05 PM -03

test-m1-edd-astro-cesteves

América do Sul (São Paulo) sa-east-1

30 Dec 2025 03:06:21 PM -03

Snapshot da conta

Informações

Visualizar painel

Atualizado diariamente

O Storage Lens fornece visibilidade sobre o uso e as tendências de atividades.

Resumo do acesso externo

Informações

Atualizado diariamente

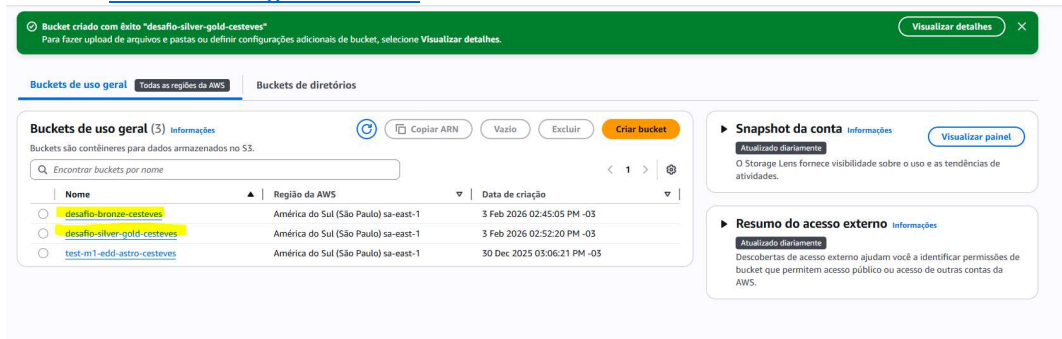
Descobertas de acesso externo ajudam você a identificar permissões de bucket que permitem acesso público ou acesso de outras contas da AWS.

PASSO 4 — Criar o Bucket S3 Silver/Gold (Data Lake):

Console AWS → Pesquise por S3 → “buckets” → “Criar bucket” → Dê nome ao bucket → Criar bucket.

Atenção: Tudo que conversa com o S3 precisa apontar para a MESMA região do bucket, neste caso: América do Sul (São Paulo) sa-east-1.

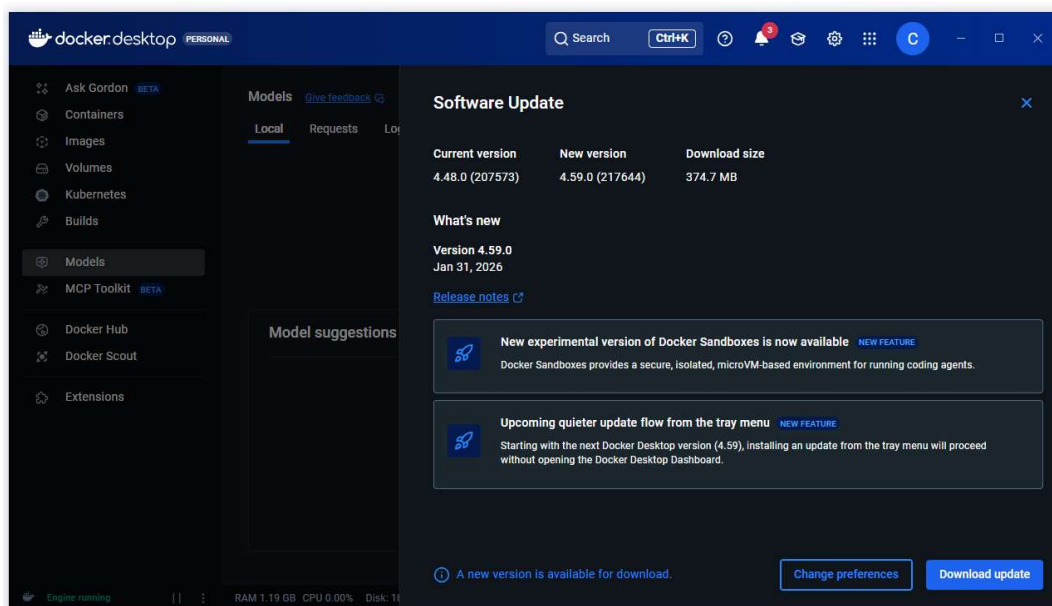
2. Buckets: [desafio-silver-gold-cesteves](#)



PASSO 5 — Baixando e configurando o Docker:

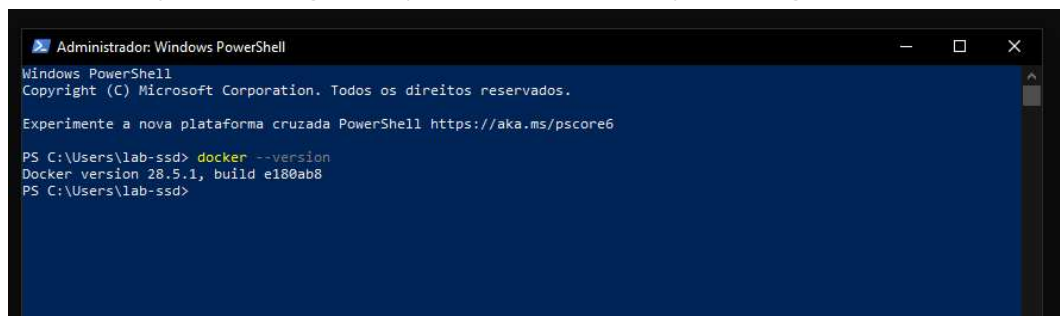
Acesse o site abaixo → “Baixar o Docker Desktop” → Execute → Marque a opção “Use WSL 2 instead of Hyper-V” (recomendado)” caso ela não esteja marcada → Reinicie a máquina.

Obs.: Já possui o Docker.



PASSO 6 — Acessando o Docker:

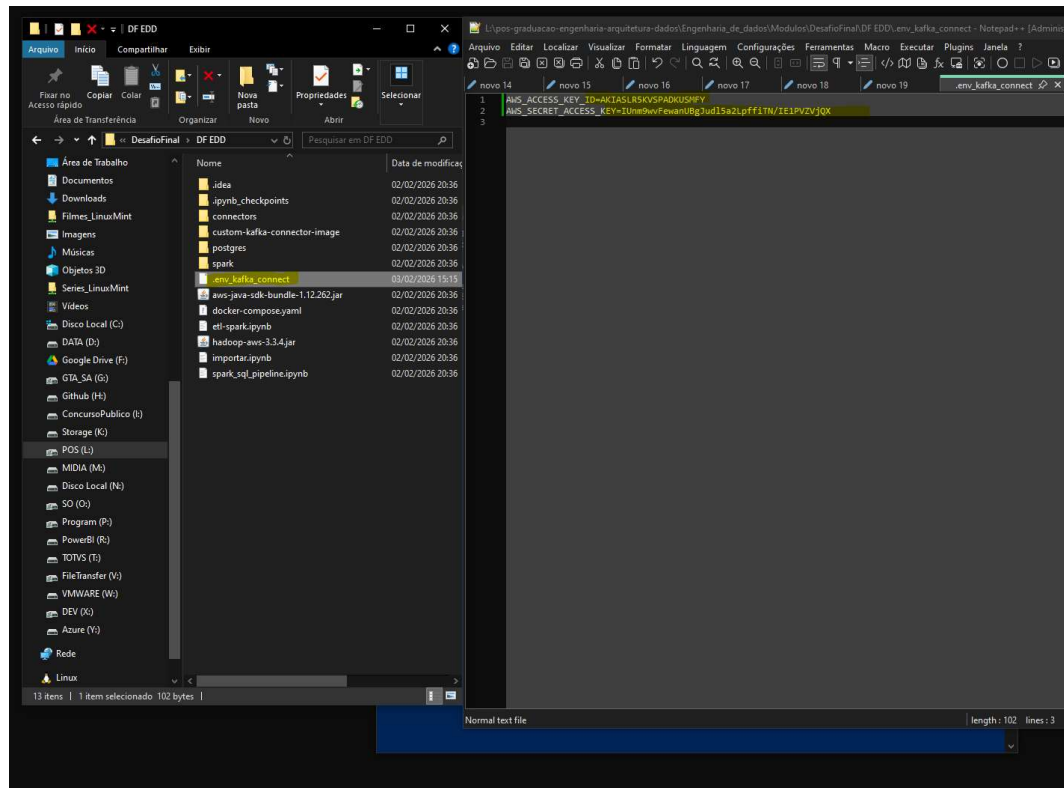
Abra o Docker Desktop → No powershell execute o comando: `docker --version` para validar que está tudo certo com a instalação feita → Aguarde aparecer “Docker Desktop is running”.



PASSO 7 — Editar o arquivo `.env_kafka_connect`:

Na raiz da pasta enviada pelo professor editado o arquivo de texto com a chave e a senha de acesso que anotamos no passo da AWS e renomeado com `.env_kafka_connect`.

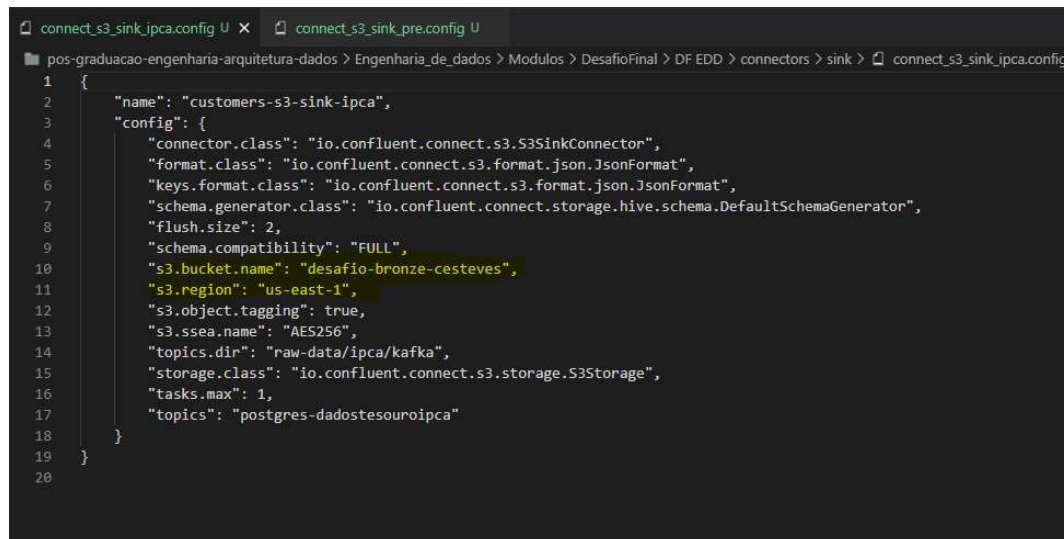
L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD



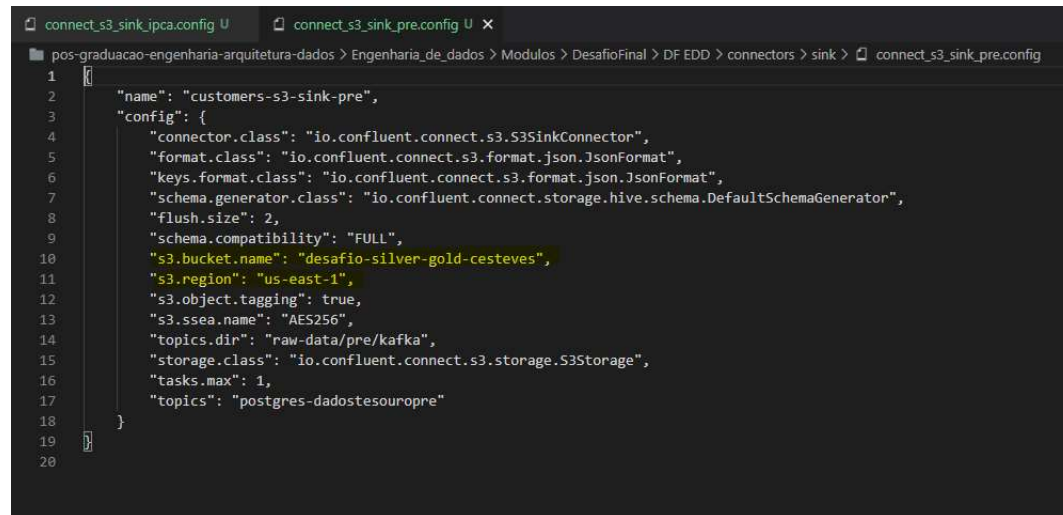
PASSO 8 — Alterar os arquivos de conectores:

L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\sink → Altere ambos os arquivos

connect_s3_sink_ipca.config



connect_s3_sink_pre.config

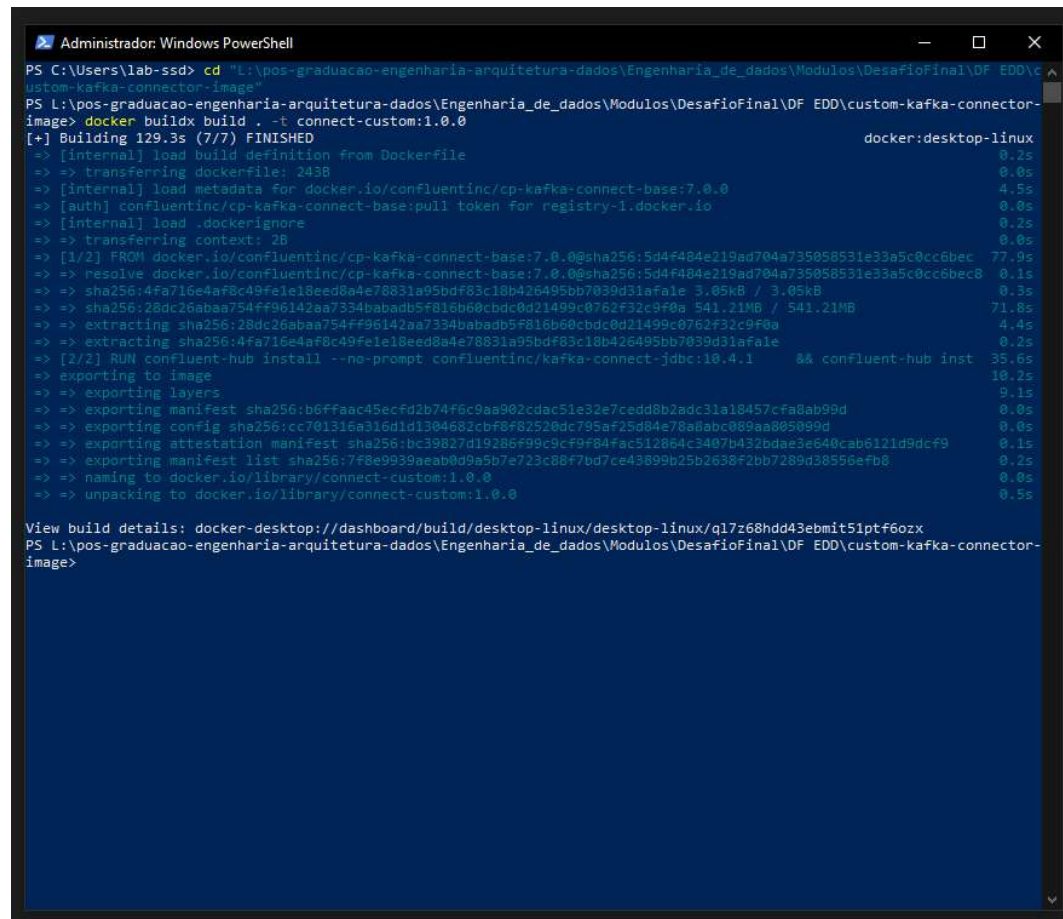


```
1  {
2    "name": "customers-s3-sink-pre",
3    "config": {
4      "connector.class": "io.confluent.connect.s3.S3SinkConnector",
5      "format.class": "io.confluent.connect.s3.format.json.JsonFormat",
6      "keys.format.class": "io.confluent.connect.s3.format.json.JsonFormat",
7      "schema.generator.class": "io.confluent.connect.storage.hive.schema.DefaultSchemaGenerator",
8      "flush.size": 2,
9      "schema.compatibility": "FULL",
10     "s3.bucket.name": "desafio-silver-gold-cesteves",
11     "s3.region": "us-east-1",
12     "s3.object.tagging": true,
13     "s3.ssea.name": "AES256",
14     "topics.dir": "raw-data/pre/kafka",
15     "storage.class": "io.confluent.connect.s3.storage.S3Storage",
16     "tasks.max": 1,
17     "topics": "postgres-dadosesouropre"
18   }
19 }
20
```

PASSO 9 — Build da imagem do Kafka Connect:

Abra o powershell e digite “cd + endereço onde está a imagem do kafka” → depois execute: docker buildx build . -t connect-custom:1.0.0

Neste desafio o caminho será: cd “L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\custom-kafka-connector-image”



```
Administrator: Windows PowerShell
PS C:\Users\lab-ssd> cd "L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\custom-kafka-connector-image"
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\custom-kafka-connector-image> docker buildx build . -t connect-custom:1.0.0
[+] Building 129.3s (7/7) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.2s
=> => transferring dockerfile: 243B                                0.0s
=> [internal] load metadata for docker.io/confluentinc/cp-kafka-connect-base:7.0.0 4.5s
=> [auth] confluentinc/cp-kafka-connect-base:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore                                  0.2s
=> => transferring context: 2B                                       0.0s
=> [1/2] FROM docker.io/confluentinc/cp-kafka-connect-base:7.0.0@sha256:5d4f484e219ad704a735058531e33a5c0cc6bec 77.9s
=> => resolve docker.io/confluentinc/cp-kafka-connect-base:7.0.0@sha256:5d4f484e219ad704a735058531e33a5c0cc6bec 0.1s
=> => sha256:4fa716e4af8c49fe1e18eed8a4e78831a95bdf83c18b426495bb7039d31afale 3.05kB / 3.05kB 0.3s
=> => sha256:26dc26abaa754ff96142aa7334babadb5f816b60cbdc0d21499c0762f32c9f0a 541.21MB / 541.21MB 71.8s
=> => extracting sha256:26dc26abaa754ff96142aa7334babadb5f816b60cbdc0d21499c0762f32c9f0a 4.4s
=> => extracting sha256:4fa716e4af8c49fe1e18eed8a4e78831a95bdf83c18b426495bb7039d31afale 0.2s
=> [2/2] RUN confluent-hub install --no-prompt confluentinc/kafka-connect-jdbc:10.4.1 && confluent-hub inst 35.6s
=> => exporting to image                                             10.2s
=> => exporting layers                                              9.1s
=> => exporting manifest sha256:b6ffaac45ecfd2b74f6c9aa902cdac51e32e7cedd8b2adc31a18457cfa8ab99d 0.0s
=> => exporting config sha256:cc701316a316d1d1304682cbf8f82520dc795af25d84e78a8abc089aa805099d 0.0s
=> => exporting attestation manifest sha256:bc39827d19286f99c9cf9f84fac512864c3407b432bdae3e640cab6121d9dcf9 0.1s
=> => exporting manifest list sha256:7f8e9939aeab0d9a5b7e723c80f7bd7ce43899b25b2638f2bb7289d38556efb0 0.2s
=> => naming to docker.io/library/connect-custom:1.0.0            0.0s
=> => unpacking to docker.io/library/connect-custom:1.0.0         0.5s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ql7z68hdd43ebmit51ptf6ozx
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\custom-kafka-connector-image>
```

PASSO 10 — Subir o PostgreSQL:

Abra o powershell e digite “cd + endereço onde está PostgreSQL” → depois execute: docker-compose up -d

Comando: cd "L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres"

Comando: docker-compose up -d

```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres> docker-compose up -d
times="2026-02-03T20:29:10-03:00" level=warning msg="L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\pos
tgres\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
time="2026-02-03T20:29:10-03:00" level=warning msg="a network with name custom_network exists but was not created for project \"postgres\".\nSet 'external:
true' to use an existing network"

[+] Container postgres started
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
6e9b969677dc   postgres:13.2                       "/docker-entrypoint.s..." 57 seconds ago Up 56 seconds   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp   postgres
bbb8c5476ef6   nginx                               "/docker-entrypoint..." 5 hours ago    Up 5 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp      meu-site
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres>
```

PASSO 11 — Instalar Jupyter Notebook e importar dados:

Abra o powershell e digite: pip install notebook → Ir para pasta do projeto → Abrir o Jupyter através do

comando: jupyter notebook → Execute o notebook de importação

Instalação: pip install notebook

```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres> pip install notebook
Collecting notebook
  Downloading notebook-7.5.3-py3-none-any.whl.metadata (10 kB)
Collecting jupyter-server<3,>=2.4.0 (from notebook)
  Downloading jupyter_server-2.17.0-py3-none-any.whl.metadata (8.5 kB)
Collecting jupyterlab-server<3,>=2.25.0 (from notebook)
  Downloading jupyterlab_server-2.28.0-py3-none-any.whl.metadata (5.9 kB)
Collecting jupyterlab<4.6,>=4.5.3 (from notebook)
  Downloading jupyterlab-4.5.3-py3-none-any.whl.metadata (16 kB)
Collecting notebook-shim<0.3,>=0.2 (from notebook)
  Downloading notebook_shim-0.2.4-py3-none-any.whl.metadata (4.0 kB)
Collecting tornado<6.2.0 (from notebook)
  Downloading tornado-6.5.4-cp39-abi3-win_amd64.whl.metadata (2.9 kB)
Collecting anyio<3.1.0 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading anyio-4.12.1-py3-none-any.whl.metadata (4.3 kB)
Collecting argon2-cffi<=21.1 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading argon2_cffi-25.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting Jinja2<3.1.6-py3-none-any.whl.metadata (2.9 kB)
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jupyter-client<7.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading jupyter_client-8.8.0-py3-none-any.whl.metadata (8.4 kB)
Collecting jupyter-core<5.0.*,>=4.12 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading jupyter_core-5.9.1-py3-none-any.whl.metadata (1.5 kB)
Collecting jupyter-events<0.11.0 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading jupyter_events-0.12.0-py3-none-any.whl.metadata (5.8 kB)
Collecting jupyter-server-terminals<0.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading jupyter_server_terminals-0.5.4-py3-none-any.whl.metadata (5.9 kB)
Collecting nbconvert<6.4.4 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading nbconvert-7.17.0-py3-none-any.whl.metadata (8.4 kB)
Collecting nbformat<5.3.0 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading nbformat-5.10.4-py3-none-any.whl.metadata (3.6 kB)
Requirement already satisfied: packaging<22.0 in c:\python314\lib\site-packages (from jupyter-server<3,>=2.4.0->notebook) (25.0)
Collecting prometheus-client<0.9 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading prometheus_client-0.24.1-py3-none-any.whl.metadata (2.1 kB)
Collecting pywinpty<2.0.1 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading pywinpty-3.0.2-cp314-cp314-win_amd64.whl.metadata (5.7 kB)
Collecting pyzmq<24 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading pyzmq-27.1.0-cp312-abi3-win_amd64.whl.metadata (6.0 kB)
Collecting sendtrush<2.1.0-py3-none-any.whl.metadata (4.1 kB)
  Downloading sendtrush-2.1.0-py3-none-any.whl.metadata (4.1 kB)
Collecting terminado<0.8.3 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading terminado-0.18.1-py3-none-any.whl.metadata (5.8 kB)
Collecting traitlets<5.6.0 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading traitlets-5.14.3-py3-none-any.whl.metadata (10 kB)
Collecting websocket-client<=1.7 (from jupyter-server<3,>=2.4.0->notebook)
  Downloading websocket_client-1.9.0-py3-none-any.whl.metadata (8.3 kB)
Collecting async-lru<1.0.0 (from jupyterlab<4.6,>=4.5.3->notebook)
  Downloading async_lru-2.1.0-py3-none-any.whl.metadata (5.3 kB)
Collecting httpx<1,>=0.25.0 (from jupyterlab<4.6,>=4.5.3->notebook)
  Downloading httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting ipykernel<6.30.0,>=6.5.0 (from jupyterlab<4.6,>=4.5.3->notebook)
  Downloading ipykernel-7.1.0-py3-none-any.whl.metadata (4.5 kB)
Collecting jupyter-lsp<2.0.0 (from jupyterlab<4.6,>=4.5.3->notebook)
  Downloading jupyter_lsp-2.3.0-py3-none-any.whl.metadata (1.8 kB)
```

Pasta do projeto:

```
Administrador: Windows PowerShell (2)
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\postgres> cd ..
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD> ls

Diretório: L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD

Mode                LastWriteTime         Length Name
----                -
d-----          02/02/2026    20:36           .idea
d-----          02/02/2026    20:36       .ipynb_checkpoints
d-----          02/02/2026    20:36       connector
d-----          02/02/2026    20:36  custom-kafka-connector-image
d-----          02/02/2026    20:36       postgres
d-----          02/02/2026    20:36         spark
-a-----          03/02/2026    15:15         102 .env_kafka_connect
-a-----          02/02/2026    20:36  280645251 aws-java-sdk-bundle-1.12.262.jar
-a-----          02/02/2026    20:36     5882 docker-compose.yml
-a-----          02/02/2026    20:36    17632 etl-spark.ipynb
-a-----          02/02/2026    20:36   962685 hadoop-aws-3.3.4.jar
-a-----          02/02/2026    20:36   103893 importar.ipynb
-a-----          02/02/2026    20:36    3016 spark_sql_pipeline.ipynb

PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD>
```

Abrindo o jupyter notebook dentro do projeto:

PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD> jupyter notebook

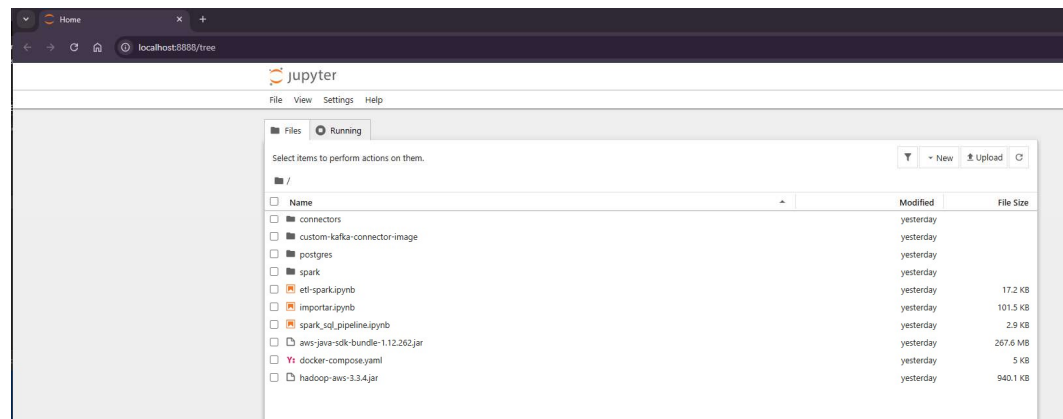
```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD> jupyter notebook
[2026-02-03 20:43:54.683 ServerApp] Extension package jupyter_lsp took 0.1950s to import
[2026-02-03 20:43:54.792 ServerApp] Extension package jupyter_server_terminals took 0.1625s to import
[2026-02-03 20:43:55.169 ServerApp] jupyter_lsp | extension was successfully linked.
[2026-02-03 20:43:55.175 ServerApp] jupyter_server_terminals | extension was successfully linked.
[2026-02-03 20:43:55.185 ServerApp] jupyterlab | extension was successfully linked.
[2026-02-03 20:43:55.193 ServerApp] notebook | extension was successfully linked.
[2026-02-03 20:43:55.198 ServerApp] Writing Jupyter server cookie secret to C:\Users\lab-ssd\AppData\Roaming\jupyter\runtime\jupyter_cookie_secret
[2026-02-03 20:43:56.082 ServerApp] notebook_shim | extension was successfully linked.
[2026-02-03 20:43:56.138 ServerApp] notebook_shim | extension was successfully loaded.
[2026-02-03 20:43:56.141 ServerApp] jupyter_lsp | extension was successfully loaded.
[2026-02-03 20:43:56.142 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[2026-02-03 20:43:56.145 LabApp] JupyterLab extension loaded from C:\Python314\Lib\site-packages\jupyterlab
[2026-02-03 20:43:56.145 LabApp] JupyterLab application directory is C:\Python314\share\jupyter\lab
[2026-02-03 20:43:56.146 LabApp] Extension Manager is 'pypl'.
[2026-02-03 20:43:56.487 ServerApp] jupyterlab | extension was successfully loaded.
[2026-02-03 20:43:56.492 ServerApp] notebook | extension was successfully loaded.
[2026-02-03 20:43:56.494 ServerApp] Serving notebooks from local directory: L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD
[2026-02-03 20:43:56.494 ServerApp] Jupyter Server 2.17.0 is running at:
[2026-02-03 20:43:56.494 ServerApp] http://localhost:8888/tree?token=cae0149a04f0908139f57650a3e13508abe9d53906601d18
[2026-02-03 20:43:56.494 ServerApp] http://127.0.0.1:8888/tree?token=cae0149a04f0908139f57650a3e13508abe9d53906601d18
[2026-02-03 20:43:56.494 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).

To access the server, open this file in a browser:
file://C:\Users\lab-ssd\AppData\Roaming\jupyter\runtime\jpservice-27664-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=cae0149a04f0908139f57650a3e13508abe9d53906601d18
http://127.0.0.1:8888/tree?token=cae0149a04f0908139f57650a3e13508abe9d53906601d18
[2026-02-03 20:43:56.544 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-languager, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-language-server-bin, yaml-language-server
```

Homepage do jupyter

<http://localhost:8888/tree>

<http://localhost:8888/tree?token=6c55d27429f3bb2a04657dbbe51795a92c83fb596e1b7aa2>



importar.ipynb

16134 rows x 9 columns

Grava os dados no Banco de Dados

```
[17]: connection_string = "postgresql://postgres:postgres@localhost:5432/postgres"
[18]: engine = create_engine(connection_string)
[19]: ipca.to_sql("dadostesouroipca", con=engine, if_exists="append", index=False)
      prefizado.to_sql("dadostesouropre", con=engine, if_exists="append", index=False)
[19]: 729
[ ]:
```

postgres rodando no docker

Containers

Container CPU usage: 0.00% / 800% (8 CPUs available)

Container memory usage: 42.36MB / 15.21GB

Name	Container ID	Image	Port(s)	CPU (%)	Actions
jovial_saha	ff3d56540408	apache/air		0%	
trusting_poincar	bb33ce286aa2	apache/air		0%	
pedantic_greide	97fee6b3a6a6	hello-world		0%	
meu-site	bbb8c547ef6	nginx	8080:80	0%	
docker	-	-	-	0%	
meu-projeto-airf	-	-	-	0%	
postgres	-	-	-	0%	
postgres	6e9b969677dc	postgres:11	5432:5432	0%	

Showing 8 items

PASSO 12 — Validar se os dados foram gravados no postgres:

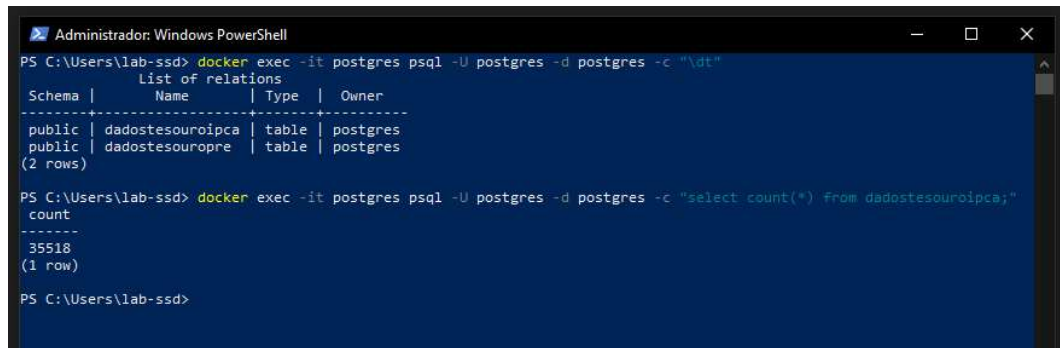
Abra o powershell e digite: `docker exec -it postgres psql -U postgres -d postgres -c "\dt"`

```
PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "\dt"
List of relations
Schema |      Name      | Type  | Owner
-----|-----|-----|-----
public | dadostesouroipca | table | postgres
public | dadostesouropre  | table | postgres
(2 rows)

PS C:\Users\lab-ssd>
```

Conte as linhas: `docker exec -it postgres psql -U postgres -d postgres -c "select count(*) from dadostesouroipca;"`

`dadostesouroipca`



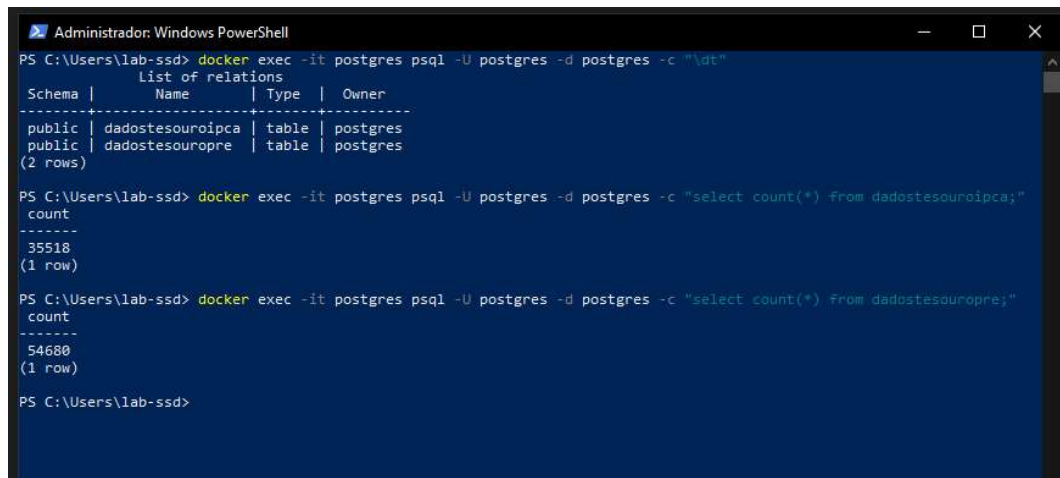
```
Administrador: Windows PowerShell
PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "\dt"
List of relations
Schema |      Name      | Type | Owner
-----+-----+-----+-----
public | dadostesouroipca | table | postgres
public | dadostesouropre | table | postgres
(2 rows)

PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "select count(*) from dadostesouroipca;"
count
-----
35518
(1 row)

PS C:\Users\lab-ssd>
```

Conte as linhas: `docker exec -it postgres psql -U postgres -d postgres -c "select count(*) from dadostesouropre;"`

`Dadostesouropre`



```
Administrador: Windows PowerShell
PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "\dt"
List of relations
Schema |      Name      | Type | Owner
-----+-----+-----+-----
public | dadostesouroipca | table | postgres
public | dadostesouropre | table | postgres
(2 rows)

PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "select count(*) from dadostesouroipca;"
count
-----
35518
(1 row)

PS C:\Users\lab-ssd> docker exec -it postgres psql -U postgres -d postgres -c "select count(*) from dadostesouropre;"
count
-----
54680
(1 row)

PS C:\Users\lab-ssd>
```

Postgres:

Dadostesouroipca

The screenshot shows the pgAdmin 4 interface with a query executed against the 'postgres/postgres@127.0.0.1' database. The query is: `SELECT "CompraManha", "VendaManha", "PUCompraManha", "PUVendaManha", "PUBaseManha", "Data_Vencimen" FROM public.dadostesouroipca;` The result set contains 21 rows of data. The columns are: `CompraManha` (double precision), `VendaManha` (double precision), `PUCompraManha` (double precision), `PUVendaManha` (double precision), `PUBaseManha` (double precision), `Data_Vencimento` (timestamp without time zone), `Data_Base` (timestamp without time zone), `Tipo` (text), and `dt_update` (timestamp without time zone). The data shows various transactions with timestamps ranging from 2015-05-15 to 2024-08-15.

	CompraManha	VendaManha	PUCompraManha	PUVendaManha	PUBaseManha	Data_Vencimento	Data_Base	Tipo	dt_update
1	6.33	6.39	1019.09	1014.58	1014.24	2015-05-15 00:00:00	2007-06-28 00:00:00	IPCA	2026-02-03 21:29:5
2	5.97	6.05	613.39	605.54	605.35	2024-08-15 00:00:00	2007-06-28 00:00:00	IPCA	2026-02-03 21:29:5
3	6.41	6.47	1012.75	1008.27	1007.93	2015-05-15 00:00:00	2007-06-27 00:00:00	IPCA	2026-02-03 21:29:5
4	6.01	6.09	609.26	601.46	601.27	2024-08-15 00:00:00	2007-06-27 00:00:00	IPCA	2026-02-03 21:29:5
5	5.97	6.05	613.01	605.16	604.96	2024-08-15 00:00:00	2007-06-26 00:00:00	IPCA	2026-02-03 21:29:5
6	6.47	6.53	1007.93	1003.47	1003.13	2015-05-15 00:00:00	2007-06-26 00:00:00	IPCA	2026-02-03 21:29:5
7	5.91	5.99	618.77	610.84	610.65	2024-08-15 00:00:00	2007-06-25 00:00:00	IPCA	2026-02-03 21:29:5
8	6.28	6.34	1021.85	1017.33	1016.99	2015-05-15 00:00:00	2007-06-25 00:00:00	IPCA	2026-02-03 21:29:5
9	6.2	6.26	1027.11	1022.56	1022.19	2015-05-15 00:00:00	2007-06-22 00:00:00	IPCA	2026-02-03 21:29:5
10	5.88	5.96	621.29	613.32	613.11	2024-08-15 00:00:00	2007-06-22 00:00:00	IPCA	2026-02-03 21:29:5
11	5.86	5.94	623.08	615.09	614.92	2024-08-15 00:00:00	2007-06-21 00:00:00	IPCA	2026-02-03 21:29:5
12	6.19	6.25	1027.51	1022.95	1022.66	2015-05-15 00:00:00	2007-06-21 00:00:00	IPCA	2026-02-03 21:29:5
13	5.85	5.93	623.92	615.92	615.75	2024-08-15 00:00:00	2007-06-20 00:00:00	IPCA	2026-02-03 21:29:5
14	6.13	6.19	1031.81	1027.22	1026.94	2015-05-15 00:00:00	2007-06-20 00:00:00	IPCA	2026-02-03 21:29:5
15	5.85	5.93	623.76	615.75	615.58	2024-08-15 00:00:00	2007-06-19 00:00:00	IPCA	2026-02-03 21:29:5
16	6.14	6.2	1030.76	1026.17	1025.89	2015-05-15 00:00:00	2007-06-19 00:00:00	IPCA	2026-02-03 21:29:5
17	5.85	5.93	623.59	615.58	615.42	2024-08-15 00:00:00	2007-06-18 00:00:00	IPCA	2026-02-03 21:29:5
18	6.15	6.21	1029.71	1025.13	1024.84	2015-05-15 00:00:00	2007-06-18 00:00:00	IPCA	2026-02-03 21:29:5
19	6.22	6.28	1024.08	1019.53	1019.16	2015-05-15 00:00:00	2007-06-15 00:00:00	IPCA	2026-02-03 21:29:5
20	5.87	5.95	621.41	613.43	613.22	2024-08-15 00:00:00	2007-06-15 00:00:00	IPCA	2026-02-03 21:29:5
21	5.9	5.98	616.19	610.26	610.06	2024-08-15 00:00:00	2007-06-14 00:00:00	IPCA	2026-02-03 21:29:5
sum	2.68	2.73	6000.52	6000.26	6000.00	2007-06-14 00:00:00	2007-06-14 00:00:00	IPCA	2026-02-03 21:29:5

Dadostesouropre

The screenshot shows the pgAdmin 4 interface with a query executed against the 'postgres/postgres@127.0.0.1' database. The query is: `SELECT "CompraManha", "VendaManha", "PUCompraManha", "PUVendaManha", "PUBaseManha", "Data_Vencimen" FROM public.dadostesouropre;` The result set contains 25 rows of data. The columns are: `CompraManha` (double precision), `VendaManha` (double precision), `PUCompraManha` (double precision), `PUVendaManha` (double precision), `PUBaseManha` (double precision), `Data_Vencimento` (timestamp without time zone), `Data_Base` (timestamp without time zone), `Tipo` (text), and `dt_update` (timestamp without time zone). The data shows various transactions with timestamps ranging from 2009-07-01 to 2024-08-15.

	CompraManha	VendaManha	PUCompraManha	PUVendaManha	PUBaseManha	Data_Vencimento	Data_Base	Tipo	dt_update
5	10.73	10.79	815.91	815.03	814.7	2009-07-01 00:00:00	2007-06-28 00:00:00	PRE-FIXAD.	2026-02-03
6	10.97	11.01	925.28	925.03	924.65	2008-04-01 00:00:00	2007-06-28 00:00:00	PRE-FIXAD.	2026-02-03
7	10.86	10.9	902.77	902.45	902.08	2008-07-01 00:00:00	2007-06-28 00:00:00	PRE-FIXAD.	2026-02-03
8	10.77	10.81	879.62	879.22	878.86	2008-10-01 00:00:00	2007-06-28 00:00:00	PRE-FIXAD.	2026-02-03
9	11.51	11.54	971.87	971.8	971.38	2007-10-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
10	11.88	11.91	999.1	999.1	998.66	2007-07-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
11	11.01	11.05	924.65	924.4	924.01	2008-04-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
12	10.92	10.96	901.92	901.59	901.22	2008-07-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
13	10.86	10.9	878.36	877.96	877.6	2008-10-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
14	10.81	10.86	855.9	855.31	854.96	2009-01-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
15	10.83	10.89	814.11	813.23	812.89	2009-07-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
16	11.24	11.28	947.33	947.15	946.75	2008-01-01 00:00:00	2007-06-27 00:00:00	PRE-FIXAD.	2026-02-03
17	11.89	11.92	998.66	998.66	998.21	2007-07-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
18	11.3	11.34	946.67	946.49	946.09	2008-01-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
19	11.57	11.6	971.31	971.24	970.81	2007-10-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
20	10.9	10.96	812.75	811.87	811.53	2009-07-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
21	11.01	11.05	900.81	900.49	900.12	2008-07-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
22	10.96	11	877	876.61	876.24	2008-10-01 00:00:00	2007-06-26 00:00:00	PRE-FIXAD.	2026-02-03
23	10.79	10.83	878.35	877.94	877.59	2008-10-01 00:00:00	2007-06-25 00:00:00	PRE-FIXAD.	2026-02-03
24	10.69	10.76	816.51	814.63	814.3	2009-07-01 00:00:00	2007-06-25 00:00:00	PRE-FIXAD.	2026-02-03
25	10.91	10.95	901.26	900.93	900.56	2008-07-01 00:00:00	2007-06-25 00:00:00	PRE-FIXAD.	2026-02-03
sum	265.45	266.45	2000.00	2000.00	2000.00	2007-06-25 00:00:00	2007-06-25 00:00:00	PRE-FIXAD.	2026-02-03

PASSO 13 — Subir a Plataforma Kafka + Kafka Connect:

Abra o powershell e digite:

cd + Raiz do projeto → Suba o ecossistema Kafka através do comando: `docker-compose -f docker-compose.yaml up -d`

Obs.: Tive um problema com o zookeeper, uma "zumbi" bloqueando o caminho, foi resolvido com o comando abaixo:

1. Remover o container conflitante (Zookeeper): `docker rm -f zookeeper`

2. Limpar possíveis conflitos dos outros serviços: `docker-compose down`
3. Subir o ambiente novamente: `O docker-compose -f docker-compose.yaml up -d`

[illegible]

Valide através do comando: `docker ps`

Neste caso sera: "L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD"

[illegible]

PASSO 14 — Criar tópicos no Kafka:

Abra o powershell e digite:

```
docker exec -it broker bash e depois digite: kafka-topics --create --bootstrap-server localhost:9092 --partitions 1
--replication-factor 1 --topic postgres-dadostesouroipca
```

```
kafka-topics --create --bootstrap-server localhost:9092 --partitions 1 --replication-factor 1 --topic postgresdadostesouopre
```

```
kafka-topics --bootstrap-server localhost:9092 --list
```

exit

[illegible]

PASSO 15 — Registrar Source Connectors (Postgres → Kafka):

Abra o powershell e digite os seguintes comandos:

Para o 550 connect_jdbc_postgres_ipca.config:

curl.exe -X POST -H "Content-Type: application/json" --data-binary "@connect_jdbc_postgres_ipca.config"
<http://localhost:8083/connectors>

Para o 544 connect_jdbc_postgres_pre.config

curl.exe -X POST -H "Content-Type: application/json" --data-binary "@connect_jdbc_postgres_pre.config"
<http://localhost:8083/connectors>

```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> curl.exe -X POST -H "Content-Type: application/json" --data-binary @connect_jdbc_postgres_pre.config http://localhost:8083/connectors
{"name":"postg-connector-ipca","config":{"connector.class":"io.confluent.connect.jdbc.JdbcSourceConnector","tasks.max":"1","connection.url":"jdbc:postgresql://postgres:5432/postgres","connection.user":"postgres","connection.password":"postgres","mode":"timestamp","timestamp.column.name":"dt_update","table.whitelist":"public.dadostesouroipca","topic.prefix":"postg-res-","validate.non.null":"false","poll.interval.ms":"500","name":"postg-connector-ipca"},"tasks":[],"type":"source"}
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> curl.exe -X POST -H "Content-Type: application/json" --data-binary @connect_jdbc_postgres_pre.config http://localhost:8083/connectors
{"name":"postg-connector","config":{"connector.class":"io.confluent.connect.jdbc.JdbcSourceConnector","tasks.max":"1","connection.url":"jdbc:postgresql://postgres:5432/postgres","connection.user":"postgres","connection.password":"postgres","mode":"timestamp","timestamp.column.name":"dt_update","table.whitelist":"public.dadostesouroipca","topic.prefix":"postg-res-","validate.non.null":"false","poll.interval.ms":"500","name":"postg-connector"},"tasks":[],"type":"source"}
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source>
```

PASSO 16 — Validar registro:

Abra o powershell e digite:

curl.exe <http://localhost:8083/connectors>

depois digite:

curl.exe http://localhost:8083/connectors/postg-connector-ipca/status

curl.exe <http://localhost:8083/connectors/postg-connector/status>

```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> curl.exe http://localhost:8083/connectors/postg-connector/status
{"name":"postg-connector","connector":{"state":"RUNNING","worker_id":"connect:8083"},"tasks":[{"id":0,"state":"RUNNING","worker_id":"connect:8083"},"type":"source"}
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> curl.exe http://localhost:8083/connectors/postg-connector-ipca/status
{"name":"postg-connector-ipca","connector":{"state":"RUNNING","worker_id":"connect:8083"},"tasks":[{"id":0,"state":"RUNNING","worker_id":"connect:8083"},"type":"source"}
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> curl.exe http://localhost:8083/connectors/postg-connector/status
{"name":"postg-connector","connector":{"state":"RUNNING","worker_id":"connect:8083"},"tasks":[{"id":0,"state":"RUNNING","worker_id":"connect:8083"},"type":"source"}
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source>
```

PASSO 17 — Confirmar que chegou dado nos tópicos (Postgres → Kafka):

Abra o powershell e digite:

docker exec -it broker bash

Depois execute:

kafka-console-consumer --bootstrap-server localhost:9092 --topic postgres-dadostesouroipca --from-beginning -
-max-messages 5

kafka-console-consumer --bootstrap-server localhost:9092 --topic postgres-dadostesouopre --from-beginning --
max-messages 5

exit

```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source> docker exec -it broker bash
[appuser@broker ~]$ kafka-console-consumer --bootstrap-server localhost:9092 --topic postgres-dadostesouroipca --from-beginning --max-messages 5
Processed a total of 5 messages
[appuser@broker ~]$ kafka-console-consumer --bootstrap-server localhost:9092 --topic postgres-dadostesouopre --from-beginning --max-messages 5
Processed a total of 5 messages
[appuser@broker ~]$ exit
exit
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\connectors\source>
```


Salvar os dados limpos no S3 em formato Parquet

silver_path = "s3a://desafio-silver-gold-cesteves/processed-data/ipca/silver/"

Salvar os dados agregados no S3 em formato Parquet

gold_path = "s3a://desafio-silver-gold-cesteves/analytics/ipca/gold/"

```
bronze_path = "s3a://desafio-bronze-cesteves/raw-data/ipca/kafka/"
|
# Ler os dados do S3 e testar a conexão
try:
    df_bronze = spark.read.json(bronze_path)
    print("Leitura bem-sucedida!")
    # Mostrar apenas os primeiros resultados e o esquema para validar
    df_bronze.printSchema()
    df_bronze.show(5)
except Exception as e:
    print(f"Erro ao acessar o S3: {e}")

# -----
# 4. Tratamento dos Dados - Camada Silver
# -----
# Remover duplicações e converter timestamps para datas legíveis
df_silver = df_bronze.dropDuplicates()

# Tratar timestamps e converter para formato legível
df_silver = df_silver.withColumn("Data_Vencimento", from_unixtime(col("Data_Vencimento") / 1000, "yyyy-MM-dd")) \
    .withColumn("Data_Base", from_unixtime(col("Data_Base") / 1000, "yyyy-MM-dd")) \
    .withColumn("dt_update", from_unixtime(col("dt_update") / 1000, "yyyy-MM-dd HH:mm:ss"))

# Tratar valores nulos
df_silver = df_silver.fillna({
    "PUCompraManha": 0,
    "PUVendaManha": 0,
    "PUBasoManha": 0
})

# Visualizar os dados transformados
print("Dados Transformados (Silver):")
df_silver.show(5, truncate=False)

# Salvar os dados limpos no S3 em formato Parquet
silver_path = "s3a://desafio-silver-gold-cesteves/processed-data/ipca/silver/"
df_silver.write.mode("overwrite").parquet(silver_path)
print("Gravado na Silver com sucesso!")

# -----
# 5. Agregação e Métricas - Camada Gold
# -----
# Calcular métricas agregadas
df_gold = df_silver.groupBy("Tipo").agg(
    avg("PUCompraManha").alias("Media_PUCompraManha"),
    avg("PUVendaManha").alias("Media_PUVendaManha"),
    count("*").alias("Total_Registros")
)

# Visualizar as métricas agregadas
print("Dados Agregados (Gold):")
df_gold.show(truncate=False)

# Salvar os dados agregados no S3 em formato Parquet
gold_path = "s3a://desafio-silver-gold-cesteves/analytics/ipca/gold/"
df_gold.write.mode("overwrite").parquet(gold_path)
print("Gravado na Gold com sucesso!")
```

PASSO 22 — Subir o Spark:

PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF
EDD\spark>

Comando:

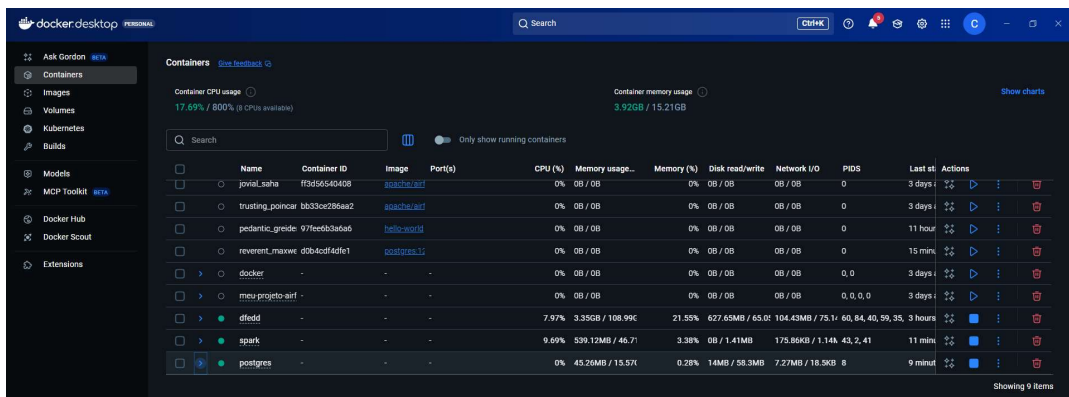
docker compose up -d

```
Administrador: Windows PowerShell

d----- 02/02/2026    20:36    jars
d----- 17/12/2024    11:12    notebooks
a----- 02/02/2026    20:36    1635 docker-compose.yaml

PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\spark> docker compose up -d
time="2026-02-04T00:49:05-03:00" level=warning msg="L:\\pos-graduacao-engenharia-arquitetura-dados\\Engenharia_de_dados\\Modulos\\DesafioFinal\\DF EDD\\spark\\docker-compose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 38/41
  spark-master Pulled
  781041b04d4fa Pull complete
  jupyter-notebook Pulled
  e98373d60b0b0 Pull complete
  acec8493d397 Pull complete
  a30f89a0a46c Pull complete
  96ee7e4439c7 Pull complete
  aa999bb9e49a Pull complete
  de2cdd097fa0 Pull complete
  f15d5d3d7ac Pull complete
  822c4cbc6a8 Pull complete
  d1526ae0cd17 Pull complete
  31873ea82d70 Pull complete
  fd92c719666c Pull complete
  2c4c69587ee4 Pull complete
  69192403e72 Pull complete
  7c128e9d20dd Pull complete
  9a165b6e9dc7 Pull complete
  80782ae10999 Pull complete
  e21a167f7127 Pull complete
  75d3399f5f2 Pull complete
  5689442f04e1 Pull complete
  8088f110b74 Pull complete
  d25166dcd7b Pull complete
  467e28fcd068 Pull complete
  85c5a5d9a5f Pull complete
  7745ee945dc Pull complete
  abaa8376a659 Pull complete
  36d4fc2e4f9f Pull complete
  7824b083412a Pull complete
  734ea8c3d94a Pull complete
  9a6a28f262a8 Pull complete
  de42adc7eb73 Pull complete
  spark-worker Pulled
  7cdbc775d228 Pull complete
  4fc7a83f9bc Pull complete
  - 52d9f6533a2 extracting 1 s
  4f4fb700ef54 Pull complete
  - 1832b0aad671 extracting 27 s
  - 396c31837116 extracting 2 s
  bb59b9756ca3 Pull complete
[+] Running 3/4
  Network spark_default Created
  Container spark-master Starting
  Container spark-worker Created
  Container jupyter-notebook Created
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint spark-master (5565c898a09a95f4285b04357b0237e3498c3cac5ac935a04a35d9a932d7c17c): Bind for 0.0.0.0:8080 failed: port is already allocated
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\spark> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
733ec6900335   confluentinc/cp-ksqldb-cli:7.0.0   "/bin/sh"               2 hours ago   Up 2 hours
e505c3b9793c   confluentinc/cp-ksqldb-server:7.0.0 "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8088->8088/tcp, [::]:8088->8088/tcp
41501b7b0fff   connect-custom:1.0.0               "/etc/confluent/dock..." 2 hours ago   Up 2 hours (healthy) 0.0.0.0:8083->8083/tcp, [::]:8083->8083/tcp
3aff93c5267d   confluentinc/cp-kafka-rest:7.0.0   "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8082->8082/tcp, [::]:8082->8082/tcp
ea989613a8a3   confluentinc/cp-schema-registry:7.0.0 "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
25bdc74d113   confluentinc/cp-kafka:7.0.0        "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:9101->9101/tcp, [::]:9101->9101/tcp
f6a87a0b345   confluentinc/cp-zookeeper:7.0.0    "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp
6e9969677dc   postgres:13.2                      "docker-entrypoint.s..." 5 hours ago   Up 5 hours   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
bbb8c5476f6   nginx                               "/docker-entrypoint..." 10 hours ago   Up 10 hours   0.0.0.0:8000->80/tcp, [::]:8000->80/tcp
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\spark>
```

Docker



Spark Master<http://localhost:8080> Ver os Workers estão vivos.

Spark Worker<http://localhost:9091> (Alterado da 8081 para 9091 para fugir do conflito).

Jupyter Notebook<http://localhost:8888> Onde ver codar.

PASSO 23 — Confirmar que o container do Jupyter do Spark subiu:

Abra o powershell e digite: docker ps

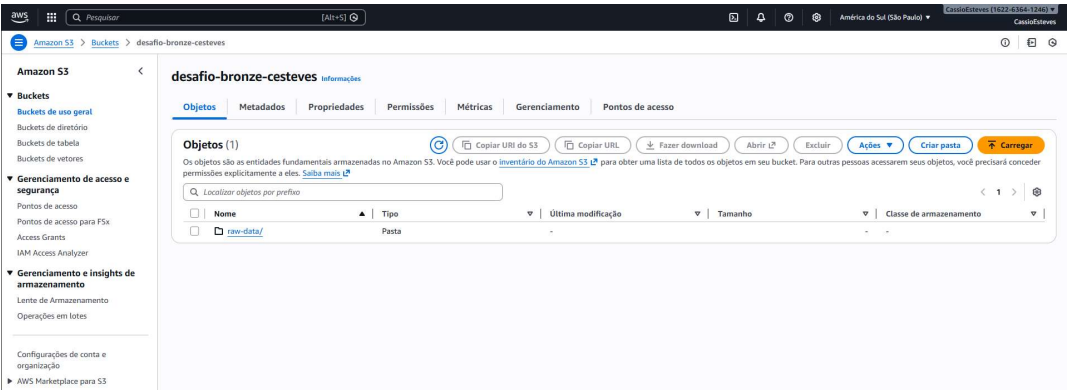
```
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\spark> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
733ec6900335   confluentinc/cp-ksqldb-cli:7.0.0   "/bin/sh"               2 hours ago   Up 2 hours
e505c3b9793c   confluentinc/cp-ksqldb-server:7.0.0 "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8088->8088/tcp, [::]:8088->8088/tcp
41501b7b0fff   connect-custom:1.0.0               "/etc/confluent/dock..." 2 hours ago   Up 2 hours (healthy) 0.0.0.0:8083->8083/tcp, [::]:8083->8083/tcp
3aff93c5267d   confluentinc/cp-kafka-rest:7.0.0   "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8082->8082/tcp, [::]:8082->8082/tcp
ea989613a8a3   confluentinc/cp-schema-registry:7.0.0 "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
25bdc74d113   confluentinc/cp-kafka:7.0.0        "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:9101->9101/tcp, [::]:9101->9101/tcp
f6a87a0b345   confluentinc/cp-zookeeper:7.0.0    "/etc/confluent/dock..." 2 hours ago   Up 2 hours   0.0.0.0:2181->2181/tcp, [::]:2181->2181/tcp
6e9969677dc   postgres:13.2                      "docker-entrypoint.s..." 5 hours ago   Up 5 hours   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
bbb8c5476f6   nginx                               "/docker-entrypoint..." 10 hours ago   Up 10 hours   0.0.0.0:8000->80/tcp, [::]:8000->80/tcp
PS L:\pos-graduacao-engenharia-arquitetura-dados\Engenharia_de_dados\Modulos\DesafioFinal\DF EDD\spark>
```


PASSO 24 — Confirmar que gravou no S3 (Entregável obrigatório):

1. Bronze

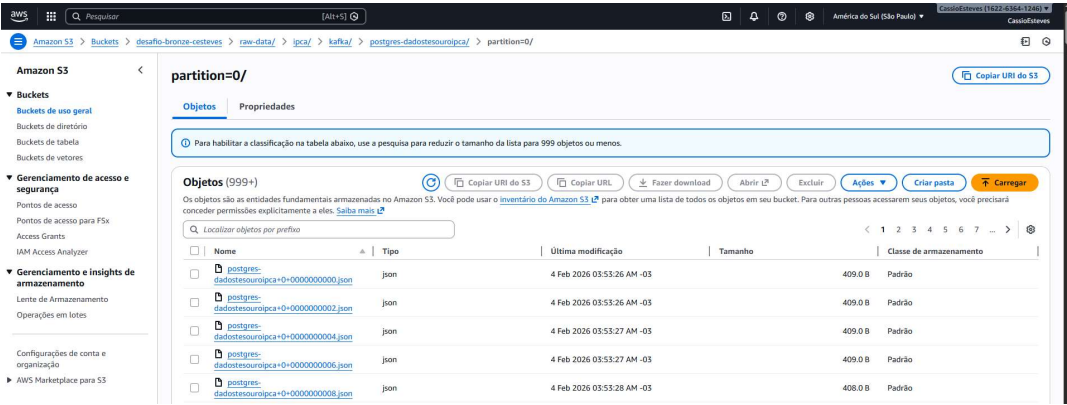
Abra [desafio-bronze-cesteves](#)

confirme a existência dos prefixos configurados

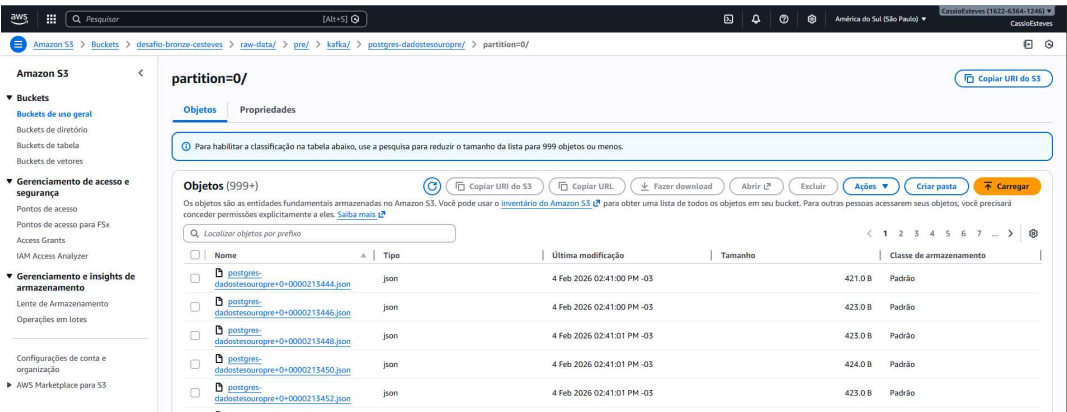


Confirmação de escrita:

Amazon S3 Buckets desafio-bronze-cesteves raw-data/ ipca/ kafka/ postgres-**dadostesouroipca/** partition=0/



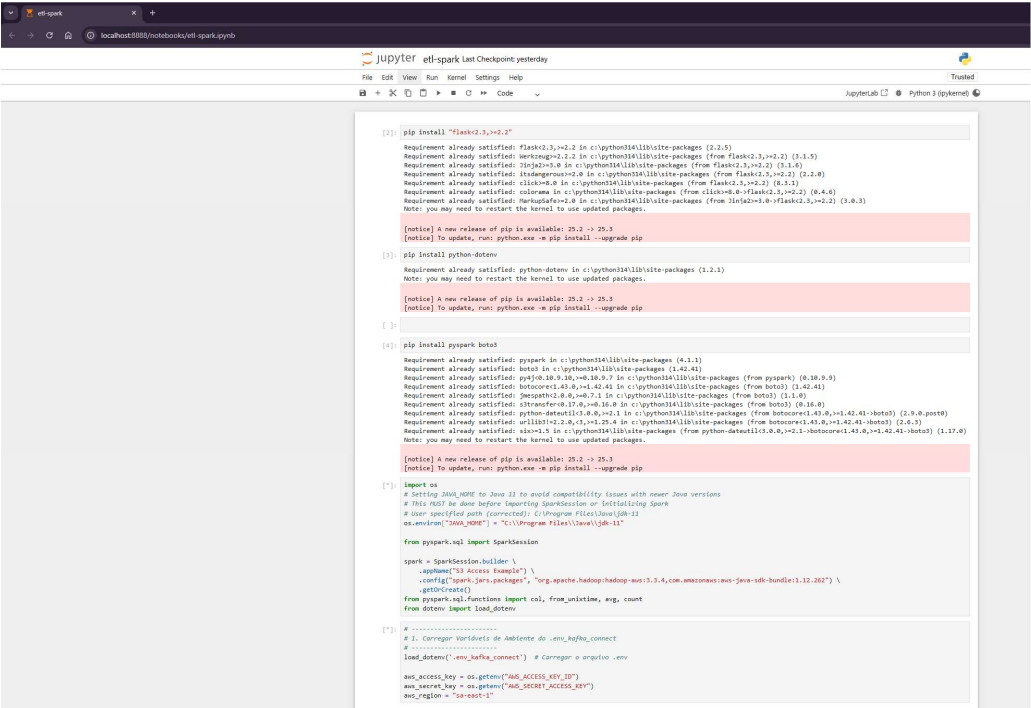
Amazon S3 Buckets desafio-bronze-cesteves raw-data/ pre/ kafka/ postgres-**dadostesouropre/** partition=0/



2. Silver

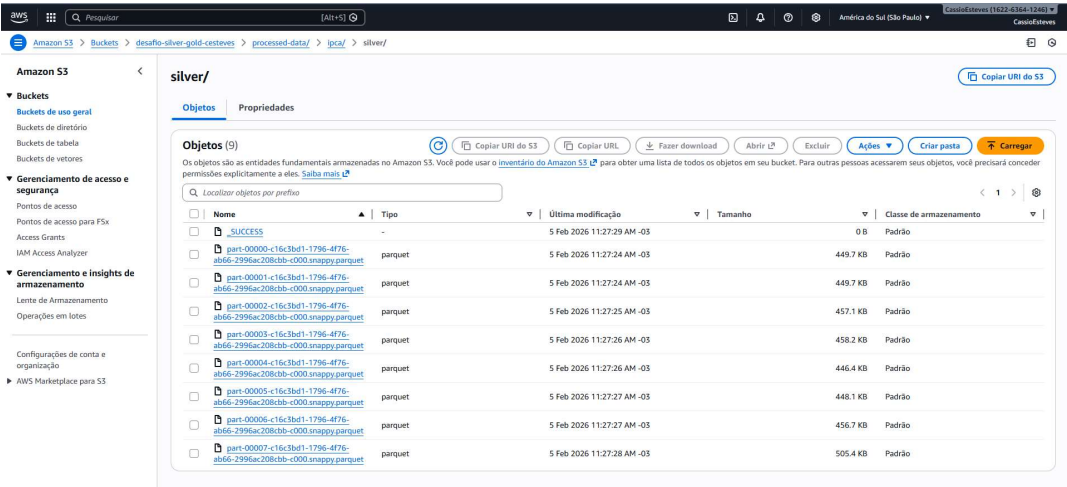
Camada SILVER — Limpeza e Padronização (Spark):

Abrir o Notebook etl-spark.ipynb → Execute



Confirmação de escrita:

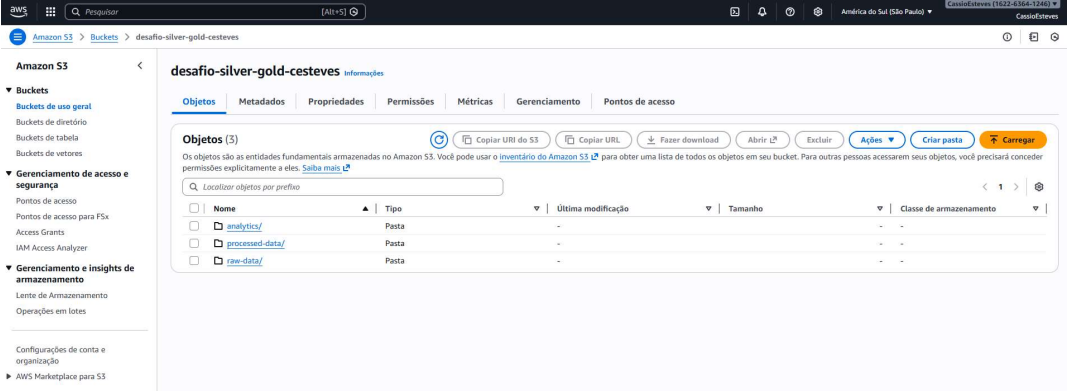
Amazon S3 Buckets desafio-silver-gold-cesteves processed-data/ ipca/ silver/



confirme a

existência dos prefixos configurados

desafio-silver-gold-cesteves



3. Gold

Amazon S3 Buckets desafio-silver-gold-cesteves analytics/ ipca/ gold/

